

STOCHASTIC METHODS IN WATER RESOURCES

Unit 6: Stochastic modelling and prediction

Lecture 20: Introduction to Kalman filter

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Kalman Filter in Hydrology: Overview

- ▶ Recursive algorithm for optimal state estimation in linear dynamic systems
- ▶ Minimizes posterior estimation error variance (Minimum Mean Square Error (MMSE) estimator)
- ▶ Combines:
 - ▶ Model prediction (hydrological process model)
 - ▶ Noisy observations (gauges, satellites)
- ▶ Typical hydrological applications:
 - ▶ Streamflow forecasting
 - ▶ Flood warning systems
 - ▶ Groundwater level estimation
 - ▶ Soil moisture assimilation
- ▶ Particularly useful when observations are sparse and uncertain

Conceptual flow:



State-Space Representation and Assumptions

System (Process) Model

$$\mathbf{x}_k = A\mathbf{x}_{k-1} + B\mathbf{u}_k + \mathbf{w}_k \quad (1)$$

Measurement Model

$$\mathbf{z}_k = H\mathbf{x}_k + \mathbf{v}_k \quad (2)$$

Definitions:

- ▶ $\mathbf{x}_k$: State vector (e.g. discharge, water table, soil moisture)
- ▶ A : State transition matrix
- ▶ H : Observation operator
- ▶ $\mathbf{w}_k \sim N(0, Q)$: Process noise
- ▶ $\mathbf{v}_k \sim N(0, R)$: Observation noise

Assumptions:

- ▶ Linearity
- ▶ Gaussian, white, zero-mean noise
- ▶ Known covariance matrices Q, R

Kalman Filter Equations (Prediction + Update)

Prediction Step

$$\hat{x}_{k|k-1} = A\hat{x}_{k-1|k-1} \quad (3)$$

$$P_{k|k-1} = AP_{k-1|k-1}A^T + Q \quad (4)$$

Update Step

$$K_k = P_{k|k-1}H^T(HP_{k|k-1}H^T + R)^{-1} \quad (5)$$

$$\hat{x}_{k|k} = \hat{x}_{k|k-1} + K_k(z_k - H\hat{x}_{k|k-1}) \quad (6)$$

$$P_{k|k} = (I - K_kH)P_{k|k-1} \quad (7)$$

Interpretation:

- ▶ K_k balances model confidence vs observation reliability
- ▶ Innovation: $z_k - H\hat{x}_{k|k-1}$ = new information

Case Study 1: Flood Routing (Muskingum Model)

State: Channel storage and discharge

Muskingum routing in state-space form:

$$Q_{k+1} = aQ_k + bI_k + w_k \quad (8)$$

Measurement:

$$z_k = Q_k + v_k \quad (9)$$

- ▶ Q_k : Channel discharge
- ▶ I_k : Inflow
- ▶ Kalman Filter corrects predicted hydrograph in real-time
- ▶ Used in flood early-warning systems

Outcome:

- ▶ Reduced peak error
- ▶ Improved lead time estimation

Simple Example: Kalman Filter for Flood Routing (Muskingum Model)

Objective: Improve flood routing discharge prediction using real-time observations.

Muskingum routing equation (discrete):

$$Q_{out}(k+1) = C_1 Q_{in}(k+1) + C_2 Q_{in}(k) + C_3 Q_{out}(k)$$

Coefficients:

$$C_1 = \frac{\Delta t - 2KX}{2K(1-X) + \Delta t}, \quad C_2 = \frac{\Delta t + 2KX}{2K(1-X) + \Delta t}, \quad C_3 = \frac{2K(1-X) - \Delta t}{2K(1-X) + \Delta t}$$

where K is the travel time of the flood wave through the reach and X is a weighting factor ($0 \leq X \leq 0.5$)

State variable:

$$x_k = Q_{out}(k)$$

State-space form:

$$x_{k+1} = C_3 x_k + C_1 u_{k+1} + C_2 u_k + w_k$$

$$z_k = x_k + v_k$$

Kalman Filter Steps:

Prediction:

$$\hat{x}_{k+1|k} = C_3 \hat{x}_{k|k} + C_1 u_{k+1} + C_2 u_k$$

$$P_{k+1|k} = C_3^2 P_{k|k} + Q$$

Update:

$$K_{k+1} = \frac{P_{k+1|k}}{P_{k+1|k} + R}$$

$$\hat{x}_{k+1|k+1} = \hat{x}_{k+1|k} + K_{k+1}(z_{k+1} - \hat{x}_{k+1|k})$$

$$P_{k+1|k+1} = (1 - K_{k+1})P_{k+1|k}$$

Simple Example: Kalman Filter for Flood Routing (Muskingum Model)

Hydrological Interpretation:

- ▶ Model propagates flood wave downstream.
- ▶ Kalman Filter corrects routing using observed hydrograph.
- ▶ Improves peak timing and magnitude prediction.
- ▶ Essential for operational flood forecasting.

Case Study 2: Groundwater Level Estimation

State Vector

$$\mathbf{x}_k = [h_k] \quad (10)$$

Groundwater dynamics (linearized form):

$$h_{k+1} = h_k + \Delta t(\alpha R_k - \beta Q_{out}) + w_k \quad (11)$$

Observation model:

$$z_k = h_k + v_k \quad (12)$$

- ▶ Assimilates piezometer readings
- ▶ Filters pumping-induced noise
- ▶ Useful in aquifer management

Benefit: Stable groundwater trend estimation under uncertainty

Case Study 3: Soil Moisture Assimilation

State: Root-zone soil moisture content

Bucket model:

$$S_{k+1} = S_k + P_k - ET_k - D_k + w_k \quad (13)$$

Observation from satellite:

$$z_k = cS_k + v_k \quad (14)$$

- ▶ Integrates remote sensing + model simulation
- ▶ Improves drought monitoring
- ▶ Basis for irrigation scheduling

Extension: EnKF preferred for distributed soil models

EKF and EnKF in Hydrology

Extended Kalman Filter (EKF)

- ▶ Nonlinear model: $x_k = f(x_{k-1}) + w_k$
- ▶ Linearization using Jacobians
- ▶ Suitable for rainfall-runoff models

Ensemble Kalman Filter (EnKF)

- ▶ Represents uncertainty using ensemble of states
- ▶ Updates each ensemble member statistically
- ▶ Widely used in distributed hydrological models

Comparison:

- ▶ EKF: Faster but local linearization errors
- ▶ EnKF: Computational but more robust

Benefits, Limitations and Conclusions

Key Benefits

- ▶ Real-time data assimilation
- ▶ Error reduction in hydrological forecasts
- ▶ Dynamic uncertainty quantification

Limitations

- ▶ Linearity assumptions
- ▶ Sensitivity to noise covariance selection
- ▶ Requires continuous observations

Conclusion

- ▶ Kalman Filter is a fundamental tool in modern hydrology
- ▶ Enables integration of physics-based models with observations
- ▶ Essential for flood, drought and groundwater management